Glossary

How to Use This Glossary

This Glossary provides the definitions of the key terms that are shown in boldface type in the text. Definitions for terms that are italicized within the text are included as well. Each glossary entry also shows the number(s) of the topic(s) where you can find the terms in their original contexts.

A

absorption a process in which energy is taken up by matter without being reflected or transmitted (3.2)

acid compound that forms H^+ ions when dissolved in water (2.4)

acid-base indicator a chemical that changes colour in response to the concentration of hydrogen ions in a solution (2.4)

activation energy the minimum amount of energy needed for a reaction to occur (2.3)

adaptation a structural or behavioural feature, or physiological process that helps an organism survive and reproduce in a particular environment (1.3)

adaptive radiation the diversification of a common ancestral species into a variety of differently adapted species (1.3)

allele a different form of the same gene (1.1)

anion a negatively charged ions (2.2)

artificial insemination a process that involves collecting and concentrating sperm, and then placing it in the female's uterus (1.4)

artificial selection selective pressure exerted by humans on populations in order to improve or modify desirable traits (1.3)

autosome all chromosomes except the sex chromosomes (1.1)

B

balanced chemical equation a complete description of a chemical reaction that provides the chemical formulas for the reactants and products and the coefficients (2.2)

base a compound that forms OH^- ions when dissolved in water (2.4)

big bang theory the theory that the universe began about 13.8 billion years ago when something unimaginably small and dense suddenly and rapidly expanded to immense size (4.4)

biotechnology the use of technology and organisms to produce useful products (1.4)

black hole the remnant of a supernova explosion with a gravitational field so strong that nothing can escape its pull (4.3)

blueshifted for objects moving toward an observer, the effect of shortening of their wavelengths toward the blue end of the visible spectrum (4.4)

C

carcinogen a substance or agent that causes cancer (1.3)

cation a positively charged ion (2.2)

celestial sphere an imaginary, rotating sphere on which lie all objects of the night sky (4.2)

cellular respiration a series of chemical reactions that release energy by reacting glucose and oxygen to produce carbon dioxide and water (3.2)

chain reaction a process in which one reaction leads to a series of further reaction (3.2)

chemical equation a representation of a chemical reaction using words or chemical formulas (2.2)

chemical mutagen a molecule that can enter the nucleus of a cell and chemically react with DNA (1.3)

chemical potential energy energy stored in chemical bonds (3.1)

chemical reaction a process in which the atoms of one or more pure substances are rearranged to form one or more different substances (2.1)

chromatin fibres of DNA in its condensed form; the usual form of DNA in the nucleus during interphase (1.1)

chromosome structures composed of DNA as a very condensed form of chromatin; visible only during cell division (1.1)

cloning a process that produces identical copies of genes, cells, or organisms (1.4)

codominance the condition in which both alleles for a trait are equally expressed in a heterozygote; both alleles are dominant (1.2)

coefficient a number placed in front of a chemical formula in a balanced chemical equation to show the ratios of substances in reaction (2.2)

combustion reaction a chemical reaction in which an element reacts with oxygen to produce an oxide of the element and heat; also refers to the burning of hydrocarbons to produce carbon dioxide and water (2.4)

complementary bases nitrogenous bases that pair together in a certain way; A and T always pair together, G and C always pair together (1.1)

conduction the transfer of thermal energy between two substances that are in direct physical contact (3.3)

constellation group of stars that form a pattern in the sky (4.2)

convection the transfer of thermal energy by the movement of heated fluids from one place to another (3.3)

cosmic microwave background (CMB) radiation radiation left over from the big bang, which fills the universe (4.4)

covalent compound a compound that results when atoms of two or more elements bond covalently (2.2)

covalent bond a strong attraction between atoms that forms when atoms share valence electrons (2.2)

cross-pollination the transfer of a male gamete from the flower of one plant to the female reproductive organ in a flower of another plant (1.2)

D

dark energy a hypothesized form of energy that makes up nearly three-quarters of the universe; has the effect of increasing the expansion of the universe (4.4)

dark matter the most abundant form of matter in the universe; invisible to telescopes (4.4)

decomposition reaction a chemical reaction in which a compound is broken down into two or more elements or simpler compounds (2.4)

DNA deoxyribonucleic acid, a double stranded nucleic acid that stores genetic information (1.1)

dominant in genetics, the allele or trait that is expressed, regardless of the identity of the other allele for the characteristic (1.2)

double replacement reaction a chemical reaction in which solutions of two ionic compounds react to produce two other ionic compounds (2.4)

E

ecliptic path the Sun and some other sky objects appear to take across the celestial sphere (4.2)

ecosystem diversity the rich diversity of ecosystems found on Earth (1.1)

elastic potential energy energy stored in a stretched or compressed object (3.1)

electrical kinetic energy energy of electrons moving form a source, along a wire or other conductor (3.1)

electrical potential energy energy stored by a separation of positive and negative charges (3.1)

electrostatic attraction the attraction between cations and anions that form the ionic bond (2.2)

endothermic reaction a chemical reaction in which there is net absorption of energy from the surroundings (2.3)

energy transfer changing the location of energy without changing its form (3.1)

energy transformation changing one form of energy to another (3.1)

exothermic reaction a chemical reaction in which there is net release of energy to the surroundings (2.3)

extinction occurs when a species completely disappears from Earth (1.3)

F

first generation plants F_1 offspring of two true-breeding plants (1.2)

formula unit the chemical formula for an ionic compound that represents smallest repeating part of the crystal lattice (2.2)

G

galaxy a collection of many billions of stars, plus gas and dust, held together by gravity (4.3)

gene a part of a chromosome that governs the expression of a trait and is passed on to offspring; it has a specific DNA base sequence (1.1)

gene cloning manipulating DNA to produce multiple copies of a gene or another segment of DNA in foreign cells (1.4)

gene pool genetic diversity within a population (1.1)

gene therapy an experimental treatment to cure genetic disorders that involves inserting a healthy, normal form of a gene into the cells of tissues that are affected by a disorder (1.4)

genetic cross any type of deliberate breeding between a genetic male and a genetic female to produce offspring that carry the genetic material of each parent (1.2)

genetic diversity the variety of inherited traits within a species (1.1)

genetics a field of biology that studies heredity, or the passing of traits from parents to offspring (1.2)

genome the complete DNA sequence in every cell of an organism (1.1)

genotype the specific combination of alleles an organism has for a trait (1.2)

gravitational potential energy energy due to the position of an object above or below another object (3.1)

H

hemophilia a recessive X-linked disorder in which blood does not clot properly (1.2)

Hertzsprung-Russell (H-R) diagram a graph that compares the properties of stars (4.3)

heterozygous an organism with two different alleles for a particular trait (1.2)

homologous chromosome a chromosome that contains the same sequence of genes as another chromosome (1.1)

homozygous an organism with two of the same alleles for a particular trait (1.2)

homozygous dominant when two identical alleles are dominant (1.2)

homozygous recessive when two identical alleles are recessive (1.2)

hybrid an offspring of parents that have different traits (1.1)

I

in vitro fertilization(IVF) a process that results in a female's eggs being fertilized by sperm outside of the body (1.4)

incomplete dominance a condition in which neither allele for a gene completely conceals the presence of the other; it results in intermediate expression of a trait (1.2)

individual a single organism (1.3)

invasive species a species that is not native to an ecosystem and causes harm to an ecosystem in which it is introduced (1.4)

ionic bond a strong attraction that forms between oppositely charged ions (2.2)

ionic compound a compound made of oppositely charged ions (2.2)

ionizing radiation radiation that is harmful to living things (3.2)

isotope two or more forms of the same element with the same number of protons but a different number of neutrons (3.2)

K

karyotype a photograph of pairs of homologous chromosomes in a cell (1.1)

kinetic energy the energy of motion (3.1)

L

law of conservation of energy the law that states that the total energy of the universe if constant; law stating that energy cannot be created or destroyed, but is transformed from one form of energy to another or transferred from one object to another (2.3, 3.1)

law of conservation of mass in a chemical reaction, the total mass of the substances used is equal to the total mass of the substances produced (2.2)

law of segregation law that describes the genetic basis for how characteristics are inherited; states that alleles for each inherited trait separate from each other during gamete formation (1.2)

light-year a unit of distance equal to the distance light travels in one year (4.3)

M

magnetic potential energy energy stored in a magnetic object because of its position in relation to a magnetic field (3.1)

main sequence a narrow band of stars on the H-R diagram that runs diagonally from the upper left (bright hot stars) to the lower right (dim cool stars) (4.3)

mechanical kinetic energy energy of motion of objects that is larger than atoms and molecules (3.1)

molecule a particle made up of two or more atoms bonded by covalent bonds (2.2)

monoculture excessive plantings of the same varieties of a species over large expanses of land (1.3)

monohybrid cross a genetic cross of parents that differ in one particular trait that is being studied (1.2)

mutagen a substance or event that increases the rate of mutation (1.3)

mutation a permanent change in the genetic material of an organism; a source of new genetic variation (1.3)

N

natural selection the process by which characteristics of a population change over many generations as organisms with heritable traits survive and reproduce, passing their traits to offspring (1.3)

neutralization reaction a chemical reaction in which an acid reacts with a base to form salt and water (2.4)

neutron star a star so dense that only neutrons can exist throughout (4.3)

nitrogenous base part of the structure of a nucleotide; nitrogenous bases in DNA are adenine (A), cytosine (C), guanine (G), and thymine (T) (1.1)

nuclear decay the change to an atom due to the emission of radiation (3.2)

nuclear energy energy stored within the nucleus of an atom (3.1)

nuclear fission a process in which a heavier nucleus is split into smaller, lighter nuclei with the release of energy (3.2)

nuclear fusion a process in which two very small nuclei combine, or fuse, to form a slightly larger nucleus (3.2)

nucleic acids DNA and RNA (ribonucleic acid) (1.1)

nucleotide consists of a phosphate group, a sugar, and a nitrogenous base; found in DNA (1.1)

P

parallax the apparent motion of a star against the background of more distant stars; used to measure the distances to and between stars (4.3)

pedigree a type of flowchart that uses symbols to show patterns of relationships and traits in a family over many generations (1.2)

pH scale a numbered scale between 0 and 14 that indicates the acidity or basicity of a solution (2.4)

phenotype the physical description of an organism's trait (1.2)

phenotypic ratio the frequency of phenotypes in offspring (1.2)

photosynthesis a series of chemical reactions that react carbon dioxide and water to produce glucose and oxygen (3.2)

physical mutagen a mutagen that causes physical changes to the structure of DNA (1.3)

plasmid a circular molecule of DNA that is commonly found in bacteria; used in genetic engineering to manufacture human insulin and other proteins (1.4)

population members of the same species living in the same geographical area at the same time (1.1)

potential energy the stored energy of an object as a result of its condition or its position (3.1)

product any new substance that is formed from a chemical reaction (2.2)

protein an organic nutrient composed of a chain of amino acids (1.1)

Punnett square a way to represent the inheritance of traits in monohybrid crosses (1.2)

R

radiant energy energy that travels as an electromagnetic wave (3.1)

radiation a mechanism of energy transfer in which atoms or molecules give off energy in the form of electromagnetic waves (3.2)

reactant a substance that undergoes a chemical change (2.2)

recessive in genetics, the allele or trait that is only expressed when two alleles are present; the expression of the allele or trait that is "hidden" or suppressed if the dominant allele is present (1.2)

recombinant DNA a DNA molecule, which includes genetic material from different sources (1.4)

redshifted for objects moving away from an observer, the effect of lengthening of their wavelengths toward the red end of the visible spectrum (4.4)

replication the process of creating an exact copy of a molecule of DNA (1.1)

retrograde motion the apparent slowing, reversal, and then looping of a planet in its path across the sky over a period of time (4.2)

S

second generation plants F_2 offspring from self-fertilization of first generation plants (1.2)

seed libraries libraries that the provides seeds and information about heritage plant varieties (1.1)

seed banks banks that preserve seeds in case of major disaster; generally not open to the public (1.1)

selective advantage a genetic advantage that improves an organism's chance of survival, usually in a changing environment (1.3)

selective pressure factors that increase or decrease the reproductive success of individuals with certain characteristics in a population (1.3)

self-fertilize when a male gamete within a flower combines with a female gamete in the same flower (1.2)

sex chromosomes the chromosomes, called X and Y, that determine the biological sex of an individual (1.1)

sex-linked traits a trait controlled by genes on sex chromosomes (1.2)

single replacement reaction a chemical reaction in which an element and a compound react to produce another element and another compound (2.4)

skeleton equation a description of a chemical reaction that provides the chemical formulas for the reactants and products (2.2)

sound energy energy of vibrations or disturbances of the particles that make up matter (3.1)

speciation the formation of new species from existing species (1.3)

species group of organisms that can interbreed in nature and produce fertile offspring (1.1)

species diversity the variety and abundance of species in a given area (1.1)

specific heat capacity the amount of energy required to change the temperature of one kilogram of a substance by one degree (3.3)

star cluster a collection of stars held together by gravity (4.3)

supernova a massive explosion in which the whole outer portion of a star is blown off (4.3)

surroundings everything else in the universe outside of the system (2.3, 3.1)

synthesis reaction a chemical reaction in which two or more reactants combine to produce a single product (2.4)

system anything that is under observation; for example, a chemical reaction (2.3, 3.1)

T

thermal energy energy of the random motion of the particles that make up matter (3.1)

traits an inherited characteristic, such as eye colour or hair colour (1.2)

transformation the process in which a cell takes up recombinant DNA during gene cloning (1.4)

transgenic organism an organism whose genetic material includes DNA from a different species (1.4)

true-breeding plants plants that consistently produce offspring with only one form of a trait (1.2)

U

universe all that exists everywhere, including all matter, energy, planets, stars, galaxies, and the space in which all of this exists (4.1)

W

word equation words that describe what happens to reactants and products in a chemical reaction (2.2)

X

X-linked traits traits controlled by genes on the X chromosome (1.2)

Index

A

T

U

V

W

X

Z

Credits

Photos

Front Matter:

xiv EHStock / iStock / Thinkstock; **xv** EHStock / iStock / Thinkstock; **xvi** Ian Crysler; **xvii** Rattiya Thongdumhyu / Shutterstock.com; **xxii** Thinkstock / jamesvancouver / iStock / Getty Images Plus; **xxiii** (top) Bill Brooks / Alamy Stock Photo, (bottom) Gunter Marx / Alamy Stock Photo; **xxiv** Chris Cheadle / Alamy Stock Photo; **xxv** Dean Werk/Great River Fishing Adventures; **xxvi** Harry Beugelink / Shutterstock.com; **xxvii** (top) Chris Cheadle / Alamy Stock Photo, (bottom) robertharding / Alamy Stock Photo

Unit 1:

2 BGSmith / Shutterstock.com; **3** (top) Eric Buermeyer / Shutterstock.com, (Centre) National Geographic Creative / Alamy Stock Photo, (bottom) Douglas Lander / Alamy Stock Photo; **4** Don W. Fawcett/Science Source; **5** (top) National Geographic Creative / Alamy Stock Photo, (centre) Chase Dekker Wild-Life Images / Getty Images, (bottom) Sean O'Neill / Alamy Stock Photo; **8–9** Don W. Fawcett/Science Source; **10** (left) Chris Cheadle / Alamy Stock Photo, (top right) Michael Wheatley / All Canada Photos, (bottom right) Design Pics Inc / Alamy Stock Photo; **11** (top, left to right) © Kristiina Paul, Barrett & MacKay / All Canada Photos, willy matheisl / Alamy Stock Photo, (centre) iStock.com/Maartje van Caspel, (inset) ASM/Science Source, (bottom) Chris Cheadle / All Canada Photos, (inset) Roy LANGSTAFF / Alamy Stock Photo; **15** Kateryna Kon / Shutterstock.com; **17** (top) Nature Picture Library / Alamy Stock Photo, (bottom left) TamasV / Shutterstock.com, (bottom right) John Zada / Alamy Stock Photo; **20** aldiami/ Andreas Alexander / Alamy Stock Photo; **21** DEPT. OF CLINICAL CYTOGENETICS, ADDENBROOKES HOSPITAL/SCIENCE PHOTO LIBRARY; **22** Kateryna Kon / Shutterstock.com; **24–25** National Geographic Creative / Alamy Stock Photo; **30** © Kristiina Paul; **33** (left) Wayne HUTCHINSON / Alamy Stock Photo, (right) Dr. Stanley Flegler/Visuals Unlimited, Inc.; **36** Andrii Zastrozhnov / Shutterstock.com; **37** Kateryna Kon / Shutterstock.com; **38** Martin Dee, Dee Creative Inc.; **41** MaximShebeko / iStock / Thinkstock; **44** Johann Schumacher / Alamy Stock Photo; **44–45** Chase Dekker Wild-Life Images / Getty Images; **45** Biophoto Associates/Science Source; **46** rainbow rays / Shutterstock.com; **47** David M. Phillips/Science Source; **48** John Zada / Alamy Stock Photo; **49** BGSmith / Shutterstock.com; **50** (left) Alan Tunnicliffe / Shutterstock.com, (right) Danita Delmont / Shutterstock.com; **52** (left) Tim Zurowski / All Canada Photos, (right) nicholas_dale / iStock / Thinkstock; **54** Radist / iStock / Thinkstock; **55** (left) Axel Bueckert / Shutterstock.com, (centre) Gladkova Svetlana / Shutterstock.com, (right) Eric Isselee / Shutterstock.com; **56** Science VU/John Doebley/ Visuals Unlimited, Inc.; **57** Mike Grandmaison / All Canada Photos; **58–59** Dave Ingram; **59** Ernie Cooper; **60** iStock.com/constantgardener; **62** iStock.com/wwing; **67** (top) Tom Reichner / Shutterstock.com, (bottom) Gualberto Becerra / Shutterstock.com; **69** Gallinago_media / Shutterstock.com; **70–71** Sean O'Neill / Alamy Stock Photo; **73** STEPHEN AUSMUS/US DEPARTMENT OF AGRICULTURE/SCIENCE PHOTO LIBRARY; **76** (left) Cultura Creative (RF) / Alamy Stock Photo, (right) Zephyr/Science Source; **78** (left) karamysh / Shutterstock.com, (centre) TFoxFoto / Shutterstock.com, (right) Wave / All Canada Photos; **79** (left) Karla Salp, Washington State Department of Agriculture, Bugwood.org, (centre) Joseph OBrien, USDA Forest Service, Bugwood.org, (right) William M. Brown Jr., Bugwood.org; **80** Image courtesy of the Non-GMO Project; **82–83** Juan Gaertner / Shutterstock.com; **84** (background) alice-photo / Shutterstock.com, (left) Matej Kastelic / Shutterstock.com, (centre) zeljkodan / Shutterstock.com, (right) Rawpixel.com / Shutterstock.com; **88** (left) Don W. Fawcett/ Science Source, (right) National Geographic Creative / Alamy Stock Photo; **89** (top) Chase Dekker Wild-Life Images / Getty Images, (bottom) Sean O'Neill / Alamy Stock Photo; **90** Gunter Marx / AG / Alamy Stock Photo; **92** (left) ddsign_stock / iStock / Thinkstock, (centre) bhofack2 / iStock / Thinkstock, (right) Diana Taliun / iStock / Thinkstock; **94** Javier Trueba/MSF/Science Source;

94–95 Sergey Nivens / Shutterstock.com; **95** (top) Design Pics Inc / Alamy Stock Photo, (bottom) NASA Goddard

Unit 2:

96–97 Ilene MacDonald / Alamy Stock Photo; **98** Lijuan Guo / Shutterstock.com; **98–99** Ilene MacDonald / Alamy Stock Photo; **99** (top) Sorbis / Shutterstock.com, (centre) Daniel Pfister, (bottom) Boomer Jerritt / All Canada Photos / Getty Images; **100–101** Ilene MacDonald / Alamy Stock Photo; **102–103** Lijuan Guo / Shutterstock.com; **104** (left) Luna Vandoorne / Shutterstock.com; (top) Aja Bariffi / Shutterstock.com, (bottom left) Tong_stocker / Shutterstock.com, (bottom right) Grandpa / Shutterstock.com; **105** (left) Africa Studio / Shutterstock.com, (top right) Bayanova Svetlana / Shutterstock.com, (bottom right) EB Adventure Photography / Shutterstock.com; **106** 2009fotofriends / Shutterstock.com; **108** (left) LanaLanglois / iStock / Thinkstock, (centre) MarkPiovesan / iStock / Thinkstock, (top right) Melinda Fawver / Shutterstock.com, (bottom right) MechanismDigital / iStock / Thinkstock; **110** (top left) Dmitry Kalinovsky / Shutterstock.com, (top right) Phonix_a / iStock / Thinkstock, (bottom left) Jewelsy / E+ / Getty Images, (bottom right) Penka Todorova Vitkova / Shutterstock.com; **111** Fotokostic / Shutterstock.com; **112–113** Charles D. Winters/Science Source; **118–119** Sorbis / Shutterstock.com; **125** Apic/RETIRED / Contributor / Hulton Archive / Getty Images; **127** ShutterPNPhotography / Shutterstock.com; **129** (all images) © Richard Megna FUNDAMENTAL PHOTOGRAPHS, NYC; **134** North Wind Picture Archives / Alamy Stock Photo; **137** Nelson Education; **138–139** Daniel Pfister; **140** Lisa F. Young / Alamy Stock Photo; **141** Stocked House Studio / Shutterstock.com, (inset) Mikael Karlsson / Alamy Stock Photo; **142** Eric Buermeyer / Shutterstock.com; **143** SpeedKingz / Shutterstock.com; **144** (left) Maurice Savage / Alamy Stock Photo, (right) Turtle Rock Scientific/Science Source; **146** schankz / Shutterstock.com, (inset) Reuter Raymond/Sygma / Getty Images; **148** Don Erhardt Photography; **149** (background) Zolnierek / Shutterstock.com, (left) Lakeview Images / Shutterstock.com, (centre) Marcin Balcerzak / Shutterstock.com, (right) A_stockphoto / Shutterstock.com; **154–155** Boomer Jerritt / All Canada Photos / Getty Images; **156** marlas111 / iStock / Thinkstock; **157** (top, series) From Silberberg, Martin, *Chemistry: The Molecular Nature of Matter and Change.* © McGraw-Hill Education/Stephen Frisch Photographer; (centre left) Laurie MacBride / Alamy Stock Photo, (centre right) THE CANADIAN PRESS/Darryl Dyck, (bottom) islandboy_stocker / Shutterstock.com; **158:** (all images) From Silberberg, Martin, *Chemistry: The Molecular Nature of Matter and Change.* © McGraw-Hill Education/Stephen Frisch Photographer; **159** (top) Junko Kimura / Staff / Getty Images, (centre) Rawpixel.com / Shutterstock.com, (bottom) Piotr Dziurman / Alamy Stock Photo; **160** (both images) Charles D. Winters/Science Source; **161** (top, series) From Silberberg, Martin, *Chemistry: The Molecular Nature of Matter and Change.* © McGraw-Hill Education/Stephen Frisch Photographer, (bottom) ER Degginger/Science Source; **162** (top left) Zelma Brezinska / Shutterstock.com, (bottom left) Todor Dobrev / Shutterstock.com, (right) PjrStudio / Alamy Stock Photo; **163** (all images) Charles D. Winters/Science Source; **164** (left) BSIP / Contributor / Universal Images Group / Getty Images, (right) ChooChin / Shutterstock.com; **166** (top) Maria Uspenskaya / Shutterstock.com, (centre) Maxvis / iStock / Thinkstock, (bottom) bogdanhoda / Shutterstock.com; **167** (top, left to right) bezikus / Shutterstock.com, Mimka / Shutterstock.com, Lyroky / Alamy Stock Photo, (centre) Danny E Hooks / Shutterstock.com, (bottom) SCIENCE PHOTO LIBRARY; **168** Peera_stockfoto / Shutterstock.com; **169** Art Directors & TRIP / Alamy Stock Photo; **171** Turtle Rock Scientific/Science Source; **172** (top left) StefanieDegner / iStock / Thinkstock, (bottom left) MarianVejcik / iStock / Thinkstock, (top right) Steve Gschmeissner/Science Source, (centre right) Conrad Elias / Alamy Stock Photo, (bottom right) IAN LINDSAY / VANCOUVER SUN. Material republished with the express permission of: Vancouver Sun, a division of Postmedia Network Inc.; **176–177** JamesChen / Shutterstock.com; **184** (top) Lijuan Guo / Shutterstock.com, (bottom) Sorbis / Shutterstock.com; **184–185** Ilene MacDonald / Alamy Stock Photo; **185** (top) Daniel Pfister, (bottom) Boomer Jerritt / All Canada

Photos / Getty Images; **186–187** Ilene MacDonald / Alamy Stock Photo; **187** Charles D. Winters/ Science Source; **188** schankz / Shutterstock.com; **188–189** Ilene MacDonald / Alamy Stock Photo; **189** khayaw / Shutterstock.com; **190** Budimir Jevtic / Shutterstock.com; **190–191** Gezim Ibrahimi / Shutterstock.com; **191** (top) Fotokostic / Shutterstock.com, (bottom) All Canada Photos / Alamy Stock Photo

Unit 3:

192–193 Rick Forgo / EyeEm / Getty Images; **194** Kateryna Kon / Shutterstock.com; **194–195** (background) Rick Forgo / EyeEm / Getty Images; **195** (top) Herschel Hoffmeyer / Shutterstock.com, (centre) NASA images / Shutterstock.com, (bottom) Global Warming Images / Alamy Stock Photo; **196–197** Rick Forgo / EyeEm / Getty Images; **198** Dennis Hallinan / Alamy Stock Photo; **198–199** Kateryna Kon / Shutterstock.com; **200** Photo by Kenneth Rivenes / Moment / Getty Images; **201** © Paul A. Souders/Corbis/VCG / Getty Images; **202** Vasily Gureev / Shutterstock.com; **203** (top left) Helene Wiesenhaan / Contributor / Getty Images, (centre left) Liliya Linnik / Shutterstock.com, (bottom left) Chris Cheadle / Alamy Stock Photo, (top right) vandame / Shutterstock.com, (bottom right) EHStock / iStock / Thinkstock; **204** (top left) Anita Patterson Peppers / Shutterstock.com, (top right) Ted Foxx / Alamy Stock Photo, (bottom) Edwin Verin / Shutterstock.com; **205** (top) TTstudio / Shutterstock.com, (bottom left) ra3rn / Shutterstock.com, (bottom right) sergign / Shutterstock.com; **208** JodiJacobson / E+ / Getty Images; **209** (left) skip caplan / Alamy Stock Photo, (right) Ascent/ PKS Media Inc. / Taxi / Getty Images; **215** (left) Johnny Adolphson / Shutterstock.com, (right) MyImages-Micha / Shutterstock.com; **216** Jeffrey Coolidge / Stone / Getty Images; **220** pixbox77 / Shutterstock.com; **220–221** Herschel Hoffmeyer / Shutterstock.com; **222** (top left) Image Source / Alamy Stock Photo, (top right) Jordan Tan / Shutterstock.com, (bottom left) ullstein bild / Contributor / Getty Images, (bottom right) Dana.S / Shutterstock.com; **224** Maxvis / iStock / Thinkstock; **225** Harry Beugelink / Shutterstock.com; **232** NASA images / Shutterstock.com; **233** TSImages / Alamy Stock Photo; **234** Studio BKK / Shutterstock.com; **236** (background) robcocquyt / Shutterstock.com, (left) Andrey_Popov / Shutterstock.com, (centre) James King-Holmes / Alamy Stock Photo, (right) Fineart1 / Shutterstock.com; **237** James MacDonald / Bloomberg / Getty Images; **238** CRISTINA QUICLER / Stringer / AFP / Getty Images; **241** Jason Payne / PNG. Material republished with the express permission of: Vancouver Sun, a division of Postmedia Network Inc.; **242** Stocktrek / Photodisc / Getty Images; **242–243** NASA images / Shutterstock.com; **244** Trong Nguyen / Shutterstock.com; **245** (top) Cultura Creative (RF) / Alamy Stock Photo, (centre) 2009fotofriends / Shutterstock.com, (bottom) Gunter Marx Photography / Corbis Documentary / Getty Images; **248** (left) Troy and Mary Parlee / Alamy Stock Photo, (centre) Chantal de Bruijne / Shutterstock.com, (right) Mauricio Handler / National Geographic / Getty Images; **249** Dietrich Leppert / Shutterstock.com; **252** (top) Peter Oshkai / Alamy Stock Photo, (bottom left) rbrown10 / Shutterstock.com, (bottom right) aquapix / Shutterstock.com; **253** (top) Pi-Lens / Shutterstock.com, (bottom left) TTshutter / iStock / Thinkstock, (bottom right) Timm Suess. This file is licensed under Creative Commons Attribution-ShareAlike 2.0 Generic (CC BY-SA 2.0) license, https://creativecommons.org/licenses/by-sa/2.0/; **254** zlikovec / Shutterstock.com; **255** poemnist / Shutterstock.com, (inset) Tory Kallman / Shutterstock.com; **256** (both images) Heiltsuk First Nation; **257** Somchai Som / Shutterstock.com; **258** David Tanaka; **261** Rawpixel.com / Shutterstock.com; **263** NatureDiver / Shutterstock.com; **264–265** Global Warming Images / Alamy Stock Photo; **265** (two images) Gary Braasch; **266** (top, left to right) Cindy Hopkins / Alamy Stock Photo, GoodWin777 / Shutterstock.com, sumnersgraphicsinc / iStock / Thinkstock, (bottom, left to right) Nadiia Kalameiets / Shutterstock.com, Patrick Durand / Contributor / Sygma / Getty Images, GuidoB. This file is licensed under the Creative Commons Attribution-Share Alike 3.0 Unported license, https://creativecommons.org/licenses/by-sa/3.0/deed.en; **267** (top) PhotoObjects.net / Thinkstock, (centre left) De Agostini Picture Library / Contributor / Getty Images, (centre right) Historical / Contributor / Corbis Historical / Getty Images, (bottom left)

Detroit Publishing Company photograph collection (Library of Congress), Prints & Photographs Division, Library of Congress, LC-DIG-det-4a04223, (bottom right) PlusONE / Shutterstock.com; **268** (top left) SolStock / E+ / Getty Images, (top right) Frances L Fruit / Shutterstock.com, (bottom) Michael Klinec / Alamy Stock Photo; **269** (top) Takashi Images / Shutterstock.com, (centre) Alison Hancock / Shutterstock.com, (bottom) EFDA-JET/SCIENCE PHOTO LIBRARY / Getty Images; **270** Bloomberg / Contributor / Getty Images; **272** wellphoto / Shutterstock.com; **273** (left) iStock.com/BrendanHunter, (right) NASA/ Glenn Benson; **275** Monkey Business Images / Shutterstock.com; **276** (left) Kateryna Kon / Shutterstock.com, (right) Herschel Hoffmeyer / Shutterstock.com; **276–277** (background) Rick Forgo / EyeEm / Getty Images; **277** (top) NASA images / Shutterstock.com, (bottom) Global Warming Images / Alamy Stock Photo; **278** Brian Caissie / Getty Images; **278–279** Rick Forgo / EyeEm / Getty Images; **280–281** (background) Rick Forgo / EyeEm / Getty Images; **282** Frederic Legrand-COMEO / Shutterstock.com; **282–283** (background) Steve Smith / Shutterstock.com; **283** (top) Radharc Images / Alamy Stock Photo, (bottom) SiliconValleyStock / Alamy Stock Photo

Unit 4:

284–285 Falkovskyi / Shutterstock.com; **286** James Wheeler / Shutterstock.com; **286–287** Falkovskyi / Shutterstock.com; **287** (top) Jess Kraft / Shutterstock.com, (centre) NASA/JPL, (bottom) Mateus A. Fandiño. This file is licensed under the Creative Commons Attribution-Share Alike 4.0 International license, https://creativecommons.org/licenses/by-sa/4.0/deed.en; **288** NASA; **292–293** James Wheeler / Shutterstock.com; **294** NASA and STScI; **295** Robert Gaines; **296** imageBROKER / Alamy Stock Photo; **297** (left) H. Armstrong Roberts/ClassicStock / Getty Images, (centre) Songkran Wannatat / Shutterstock.com, (right) crazystocker / Shutterstock.com; **298** urbans / Shutterstock.com; **299** Jukka Risikko / Shutterstock.com; **300–301** Jess Kraft / Shutterstock.com; **302** James Wheeler / Shutterstock.com; **303** (top) David Hajnal / Shutterstock.com, (centre) Shaun Cunningham / Alamy Stock Photo, (bottom) John R. Foster/Science Source; **304** (top) Daderot, (centre) Dominic Robinson / Alamy Stock Photo, (bottom) 1000 Words / Shutterstock.com; **305** (top) PA Images / Alamy Stock Photo, (bottom) Universal History Archive / Contributor / Universal Images Group Editorial / Getty Images; **306** (both images) Eckhard Slawick/Photo Researchers, Inc.; **308** Stocktrek Images, Inc. / Alamy Stock Photo; **310** (bottom, two images) Dave Starrett; **311** (all images) Dave Starrett; **312** (both images) Dave Starrett; **314–315** Maxim Tupikov / Shutterstock.com; **318** (both images) Dave Starrett; **319** (all images) Dave Starrett; **320** NASA/JPL; **321** NASA/JPL; **322** Max Lindenthaler / Shutterstock.com; **323** (top left) NASA/JPL, (top right) NASA/JPL/STScI, (bottom left) NASA/JPL/STScI, (bottom right) I. de Pater, J.R. Dickel; NRAO/AUI/NSF; **324** © Canadian Space Agency, 2002; **324–325** Zakharchuk / Shutterstock.com; **325** © Canadian Space Agency, 2002; **326** JamesChen / Shutterstock.com; **327** (top left) NASA, ESA and the Hubble Heritage (STScI/AURA)-ESA/Hubble Collaboration. Acknowledgment: M. West (ESO, Chile), (top right) NASA/JPL-Caltech/ESA/Harvard-Smithsonian CfA, (bottom left) NASA Goddard, (bottom right) NASA Goddard; **329** (left) Giovanni Benintende / Shutterstock.com, (right) NASA Goddard; **330** (top) ESO/NASA/JPL-Caltech/M. Kornmesser/ R. Hurt, (bottom) MaIII Themd / Shutterstock.com; **332** NASA/JPL/STScI Hubble Deep Field Team; **333** (background) Belish / Shutterstock.com, (left) David Parker/Science Source, (centre) Hero Images Inc. / Alamy Stock Photo, (right) schlyx / Shutterstock.com; **335** (both images) Dave Starrett; **339** Genevieve de Messieres / Shutterstock.com; **342** Tragoolchitr Jittasaiyapan / Shutterstock.com; **344** NASA/MSFC; **345** (both images) © Australian Astronomical Observatory, photographs by David Malin; **346** NASA Goddard; **349** NASA, ESA, and the Hubble Heritage Team (STScI/AURA)-ESA/Hubble Collaboration; Acknowledgment: B. Whitmore (Space Telescope Science Institute); **350** Jurik Peter / Shutterstock.com; **350–351** Thinkstock / Aphelleon / iStock / Getty Images Plus; **351** Daily Herald Archive / SSPL / Getty Images; **352–353** Mateus A. Fandiño. This file is licensed under the Creative Commons Attribution-Share Alike 4.0 International license, https://creativecommons.org/

licenses/by-sa/4.0/deed.en; **354** Pictorial Press Ltd / Alamy Stock Photo; **357:** Dave Starrett; **358** (both images) NASA / WMAP Science Team; **359** The World in HDR / Shutterstock.com; **360–361** (background) Egyptian Studio / Shutterstock.com; **362** (both images) NASA, ESA, M.J. Jee and H. Ford (Johns Hopkins University); **363** (top) NASA/JPL-Caltech, (bottom) ESO/L. Calçada; **364** NASA/ESA, The Hubble Key Project Team and The High-Z Supernova Search Team; **366** Spencer Sutton / Science Source; **368** Bruce Rolff / Shutterstock.com; **369** ESA and the Planck Collaboration; **370** (left) James Wheeler / Shutterstock.com, (right) Jess Kraft / Shutterstock.com; **370–371** Falkovskyi / Shutterstock.com; **371** (top) NASA/JPL, (bottom) Mateus A. Fandiño. This file is licensed under the Creative Commons Attribution-Share Alike 4.0 International license, https://creativecommons.org/licenses/by-sa/4.0/deed.en; **372** Thinkstock / yoko_ken_chan / iStock / Getty Images Plus; **372–373** Falkovskyi / Shutterstock.com; **373** Allexxandar / Shutterstock.com; **374–375** Falkovskyi / Shutterstock.com; **375** David Coleman / Alamy Stock Photo; **376** (top) NASA/Chris Gun, (bottom) jiawangkun / Shutterstock.com; **376–377** NASA; **377** Falkovskyi / Shutterstock.com

Back Matter:

382 (three images) From *Science Connections 9*. Nelson Education; **391** (left) © Heather Rushton | Dreamstime.com, (right) iStock.com/track5; **392** (all images) iStock.com/track5; **394** Don Hammond/Design Pics; **395** Andersen Ross/Blend Images; **396** Fuse / Corbis / Getty Images; **397** iStock.com/alvarez; **399** (left) Don Hammond/Design Pics, (top right) Gilles Mingasson/Hulton Archive/Getty Images, (bottom right) Akhararat Wathanasing / Alamy Stock Photo

Art

Unit 2:

101 From SANADER/DAVIES/REITER/SALCICCIOLI. *CHEMISTRY 11/12 STUDENT RESOURCE*, 1E. © 2019 Nelson Education Ltd. Reproduced by permission. www.cengage.com/permissions; **108** Workplace Hazardous Materials Information System (WHMIS) pictograms, http://www.ccohs.ca/WHMISpictograms.html, Canadian Centre for Occupational Health and Safety (CCOHS), 2015. Reproduced with the permission of CCOHS, 2018; **186** (Rev. 2 (a) and (c)) From Di Giuseppe. *Nelson Science Connections 10*, 1E. © 2010 Nelson Education Ltd. Reproduced by permission. www.cengage.com/permissions

Unit 3:

238 From Barker/Fraser/Davies/Vucic. *Nelson Science Perspectives* 9, 1E. © 2010 Nelson Education Ltd. Reproduced by permission. www.cengage.com/permissions

Unit 4:

288 From Barker/Fraser/Davies/Vucic. *Nelson Science Perspectives* 9, 1E. © 2010 Nelson Education Ltd. Reproduced by permission. www.cengage.com/permissions; **306** (both overlain pieces of art) From Barker/Fraser/Davies/Vucic. *Nelson Science Perspectives* 9, 1E. © 2010 Nelson Education Ltd. Reproduced by permission. www.cengage.com/permissions; **308** From Barker/Fraser/Davies/Vucic. *Nelson Science Perspectives* 9, 1E. © 2010 Nelson Education Ltd. Reproduced by permission. www.cengage.com/permissions; **309** (top and bottom) From Barker/Fraser/Davies/Vucic. *Nelson Science Perspectives* 9, 1E. © 2010 Nelson Education Ltd. Reproduced by permission. www.cengage.com/permissions **311** From Barker/Fraser/Davies/Vucic. *Nelson Science Perspectives* 9, 1E. © 2010 Nelson Education Ltd. Reproduced by permission. www.cengage.com/permissions; **316** From Barker/Fraser/Davies/Vucic. *Nelson Science Perspectives* 9, 1E. © 2010 Nelson Education Ltd. Reproduced by permission. www.cengage.com/permissions; **316** (top, centre, bottom) From Barker/Fraser/Davies/Vucic. *Nelson Science Perspectives* 9, 1E. © 2010 Nelson Education Ltd. Reproduced by permission. www.cengage.com/permissions; **317** From Barker/Fraser/Davies/Vucic. *Nelson Science Perspectives* 9, 1E. © 2010 Nelson Education Ltd. Reproduced by permission. www.cengage.com/permissions; **328** From Barker/Fraser/Davies/Vucic. *Nelson Science Perspectives* 9, 1E. © 2010 Nelson Education Ltd. Reproduced by permission. www.cengage.com/permissions; **331** From Barker/Fraser/Davies/Vucic. *Nelson Science Perspectives* 9, 1E. © 2010 Nelson Education Ltd. Reproduced by permission. www.cengage.com/permissions; **336** (top and bottom) From Barker/Fraser/Davies/Vucic. *Nelson Science Perspectives* 9, 1E. © 2010 Nelson Education Ltd. Reproduced by permission. www.cengage.com/permissions; **341** From Barker/Fraser/Davies/Vucic. *Nelson Science Perspectives* 9, 1E. © 2010 Nelson Education Ltd. Reproduced by permission. www.cengage.com/permissions; **343** From Barker/Fraser/Davies/Vucic. *Nelson Science Perspectives* 9, 1E. © 2010 Nelson Education Ltd. Reproduced by permission. www.cengage.com/permissions; **347** From Barker/Fraser/Davies/Vucic. *Nelson Science Perspectives* 9, 1E. © 2010 Nelson Education Ltd. Reproduced by permission. www.cengage.com/permissions

Back Matter:

411 From Barker/Fraser/Davies/Vucic. *Nelson Science Perspectives* 9, 1E. © 2010 Nelson Education Ltd. Reproduced by permission. www.cengage.com/permissions